AQ11.3: Activity Questions 3 - Not Graded

This assignment will not be graded and is only for practice.

Level 1

Rank of the Hessian matrix $H_f(x, y, z)$Hf(x, y, z) of the function
$f(x, y, z) = 3x^2 + xz + 2zy - z^2$ f(x, y, z) = 3x2 + xz + 2zy − z2 at point $(1, 0, 1)$(1, 0, 1) is

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 3
1 point
1 point
Choose the correct statements about $f(x, y, z) = 3x^2z + 2z - 4zy^2$ f(x, y, z) = 3x2z + 2z − 4zy2.

☐ $(0, 0, 0)$(0, 0, 0) is a critical point of f f.

☐ Hessian matrix of f f at point (x, y, z)(x, y, z) is $\begin{bmatrix} 6z & 0 & 6x \\ 0 & -8z & -8y \\ 6x & -8y & 0 \end{bmatrix}$ 6z06x0−8z−8y6x−8y0 .

☐ Hessian matrix of f f at point (x, y, z)(x, y, z) is $\begin{bmatrix} 6z & 0 & 6x \\ 0 & -8y & -8z \\ 6x & -8z & -8y \end{bmatrix}$ 6z06x0−8y−8z6x−8z−8y .

☐ Rank of the Hessian matrix of f f at point $(3, -1, 2)$(3, −1, 2) is 33.

☐ Rank of the Hessian matrix of f f at point $(3, -1, 2)$(3, −1, 2) is 22.

No, the answer is incorrect.
Score: 0
Accepted Answers:

Hessian matrix of f f at point (x, y, z)(x, y, z) is $\begin{bmatrix} 6z & 0 & 6x \\ 0 & -8z & -8y \\ 6x & -8y & 0 \end{bmatrix}$ 6z06x0−8z−8y6x−8y0 .

Rank of the Hessian matrix of f f at point $(3, -1, 2)$(3, −1, 2) is 33.

Consider the following function

$$f(x, y, z) = x^3 + xy^2 + x^2 + y^2 + 3z^2 \quad \text{f(x, y, z) = x3 + xy2 + x2 + y2 + 3z2}$$

Answer questions 3,4,5 and 6 based on the function f f.
Number of critical points of f f is

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point
Rank of the Hessian matrix of f f at point $(\frac{-1}{3}, 0, 0)(3{-}1, 0, 0)$ is

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point
Rank of the Hessian matrix of f f at point $(0, 0, 0)(0, 0, 0)$ is

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 3
1 point

Level 2

1 point
Choose the correct statements.

☐ $(\frac{-1}{3}, 0, 0)(3{-}1, 0, 0)$ is a saddle point of f f.

☐ $(\frac{-2}{3}, 0, 0)(3{-}2, 0, 0)$ is a saddle point of f f.

☐ $(\frac{-2}{3}, 0, 0)(3{-}2, 0, 0)$ is a local minimum of f f.

☐ $(0, 0, 0)(0, 0, 0)$ is a local minimum of f f.

☐ $(0, 0, 0)(0, 0, 0)$ is a local maximum of f f.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$(\frac{-2}{3}, 0, 0)(3{-}2, 0, 0)$ is a saddle point of f f.
$(0, 0, 0)(0, 0, 0)$ is a local minimum of f f.
Let aa denote the shortest distance from the origin to the plane $z = x + y + 3$z = x + y + 3. The value of a^2 a2 is

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 3
1 point
1 point
Consider a function $f(x, y, z)$f(x, y, z). Let v_1, v_2, v_3 v1, v2, v3 be the critical points of the function f f and H_k Hk be the Hessian matrix of the function at the point v_k vk.Then which of the following options is(are) true?

☐ If $H_1 = \begin{bmatrix} -1 & 0 & 0 \\ 1 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix}$ H1 = $\begin{bmatrix} & & \\ & & \\ -1 & 1 & 10 & 1 & -1 & 1 \end{bmatrix}$, then v_1 v1 is a saddle point.

☐ If $H_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ H2 = $\begin{bmatrix} & & \\ & & \\ 1 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 3 \end{bmatrix}$, then v_2 v2 is a saddle point.

☐ If $H_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ H3 = $\begin{bmatrix} & & \\ & & \\ 1 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 3 \end{bmatrix}$, then v_3 v3 is a point of local maximum.

☐ If $H_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ H1 = $\begin{bmatrix} & & \\ & & \\ 1 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 3 \end{bmatrix}$, then v_1 v1 is a point of local minimum.

☐ If $H_2 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix}$ H2 = $\begin{bmatrix} & & \\ & & \\ -1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 & -3 \end{bmatrix}$, then v_2 v2 is a point of local maximum.

☐ If $H_3 = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ H3 = $\begin{bmatrix} & & \\ & & \\ -2 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 \end{bmatrix}$, then v_3 v3 is a point of local minimum.

No, the answer is incorrect.
Score: 0
Accepted Answers:

If $H_1 = \begin{bmatrix} -1 & 0 & 0 \\ 1 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix}$ H1 = $\begin{bmatrix} & & \\ & & \\ -1 & 1 & 10 & 1 & -1 & 1 \end{bmatrix}$, then v_1 v1 is a saddle point.

If $H_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ H1 = $\begin{bmatrix} & & \\ & & \\ 1 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 3 \end{bmatrix}$, then v_1 v1 is a point of local minimum.

If $H_2 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix}$ H2 = $\begin{bmatrix} & & \\ & & \\ -1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 & -3 \end{bmatrix}$, then v_2 v2 is a point of local maximum.

Check Answers